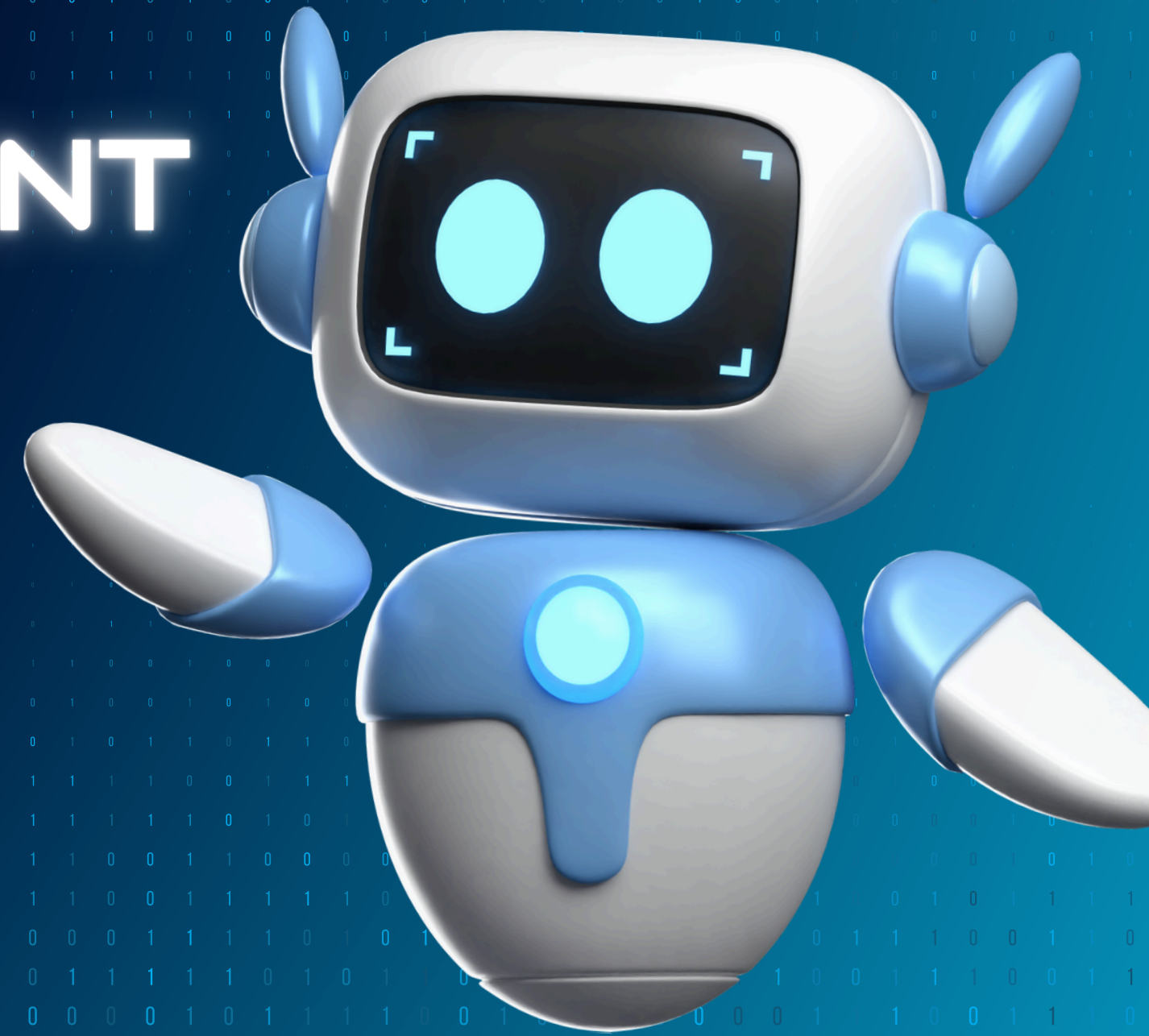


IA

INSTRUCTOR ASSISTANT

Interactive AI Assistant with
Voice and Text Integration



CONTENTS

- ① Project Overview
- ② System Architecture
- ③ Backend Details
- ④ Moderation
- ⑤ AI Integration
- ⑥ Demo
- ⑦ Challenges & Solutions
- ⑧ Future Enhancements

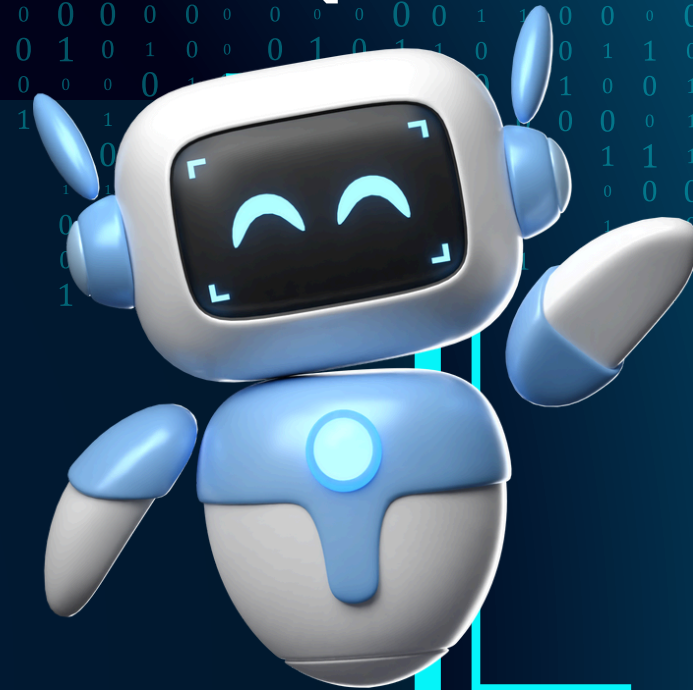
- ⑨ Conclusion



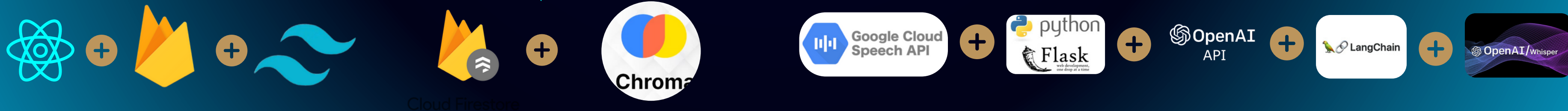


PROJECT OVERVIEW

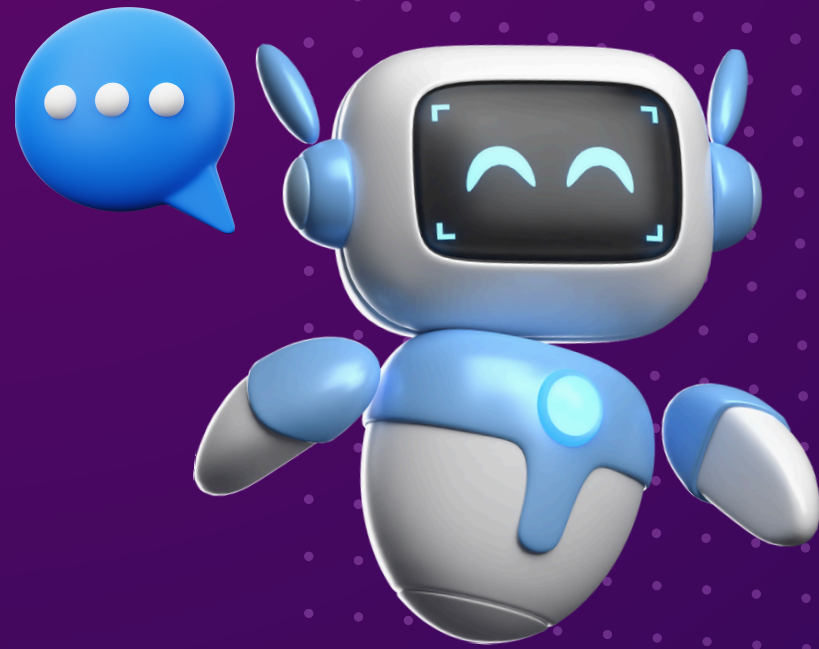
Create a dynamic chatbot using React (frontend) and Flask (backend).



- Features:
 - Text-based chat and voice interaction.
 - File upload and web link embedding.
 - YouTube transcript processing.
- Technology Stack:
 - Frontend: React, Tailwind CSS, Firebase.
 - Backend: Flask, OpenAI APIs, LangChain, Whisper, Google Cloud Speech API.
 - Database: Firestore, Chroma.



② SYSTEM ARCHITECTURE



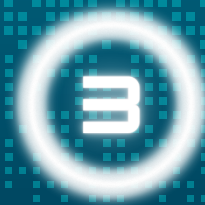
USERS INTERACT VIA
TEXT/VOICE.

FRONTEND (REACT) HANDLES
INPUT/OUTPUT.

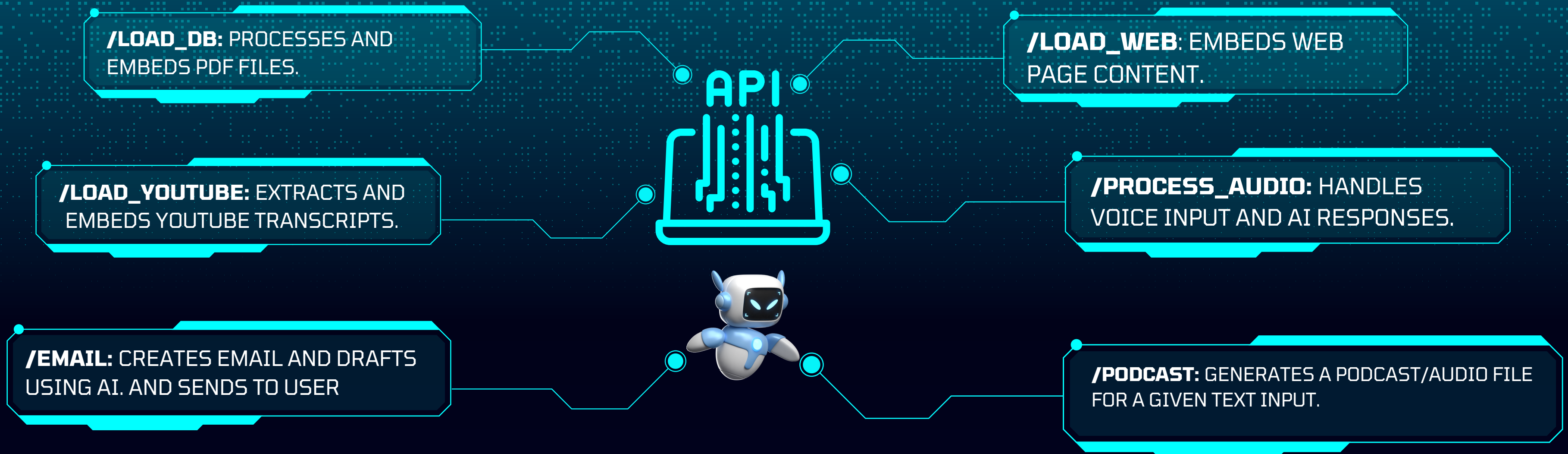
BACKEND (FLASK): PROCESSES REQUESTS
(FILES, URLS, AUDIO) AND COMMUNICATES
WITH OPENAI APIS.

RESPONSES DISPLAYED ON THE FRONTEND.

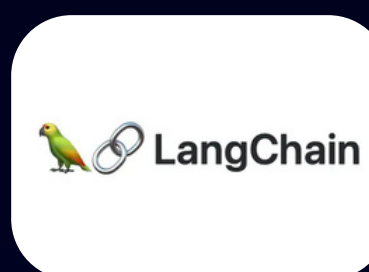
AI PROCESSING WITH LANGCHAIN AND
WHISPER.



BACKEND DETAILS



Libraries: Flask, LangChain, Whisper, Pydub.



4 MODERATION SYSTEM: SAFEGUARDING INPUTS

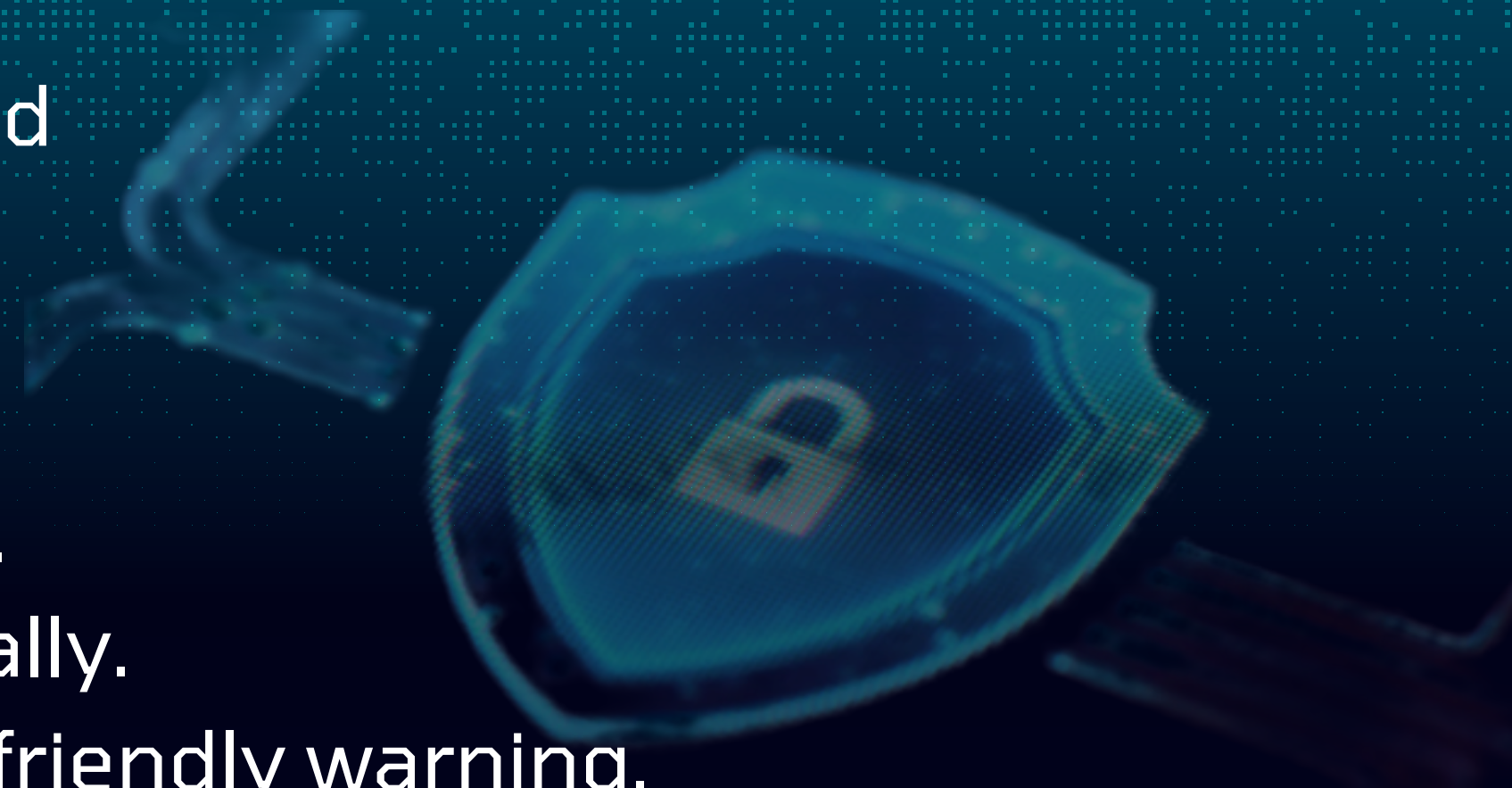
Ensure all user inputs are appropriate and safe for everyone.

Input Check:

- The system reviews what users type.
- Safe messages are processed normally.
- Unsafe messages are flagged with a friendly warning.

Flagged Message Response:

- Users see a helpful message asking them to revise their input.
- Problematic messages are saved securely for review.



⑤ AI INTEGRATION

OPENAI MODELS:

- GPT-3.5/4 for conversational responses.
- Whisper for voice transcription.
- Used about 987560 trained tokens for Fine Tuning

LANGCHAIN:

- Document processing, embedding, and retrieval.

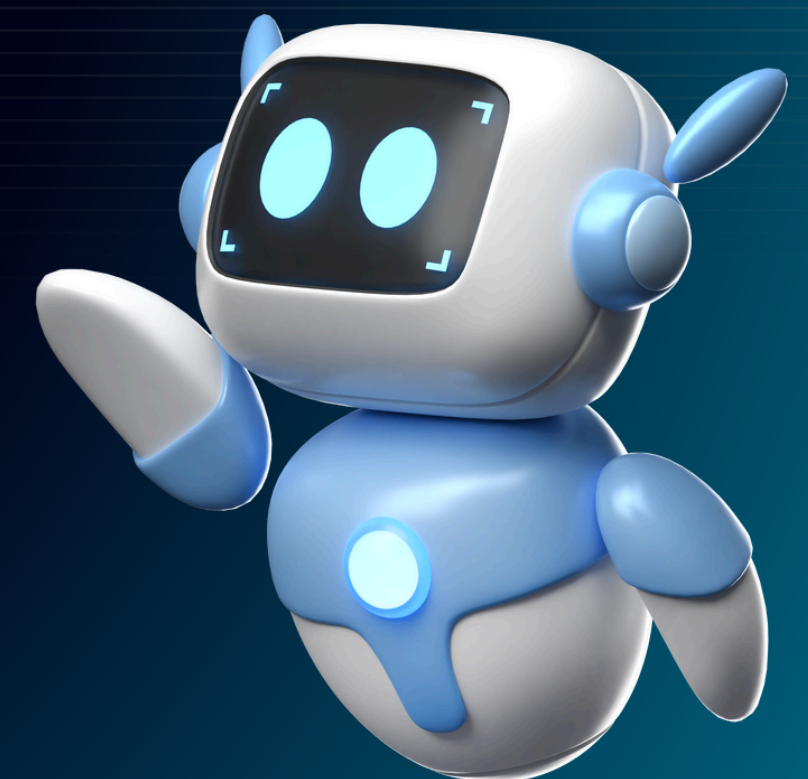
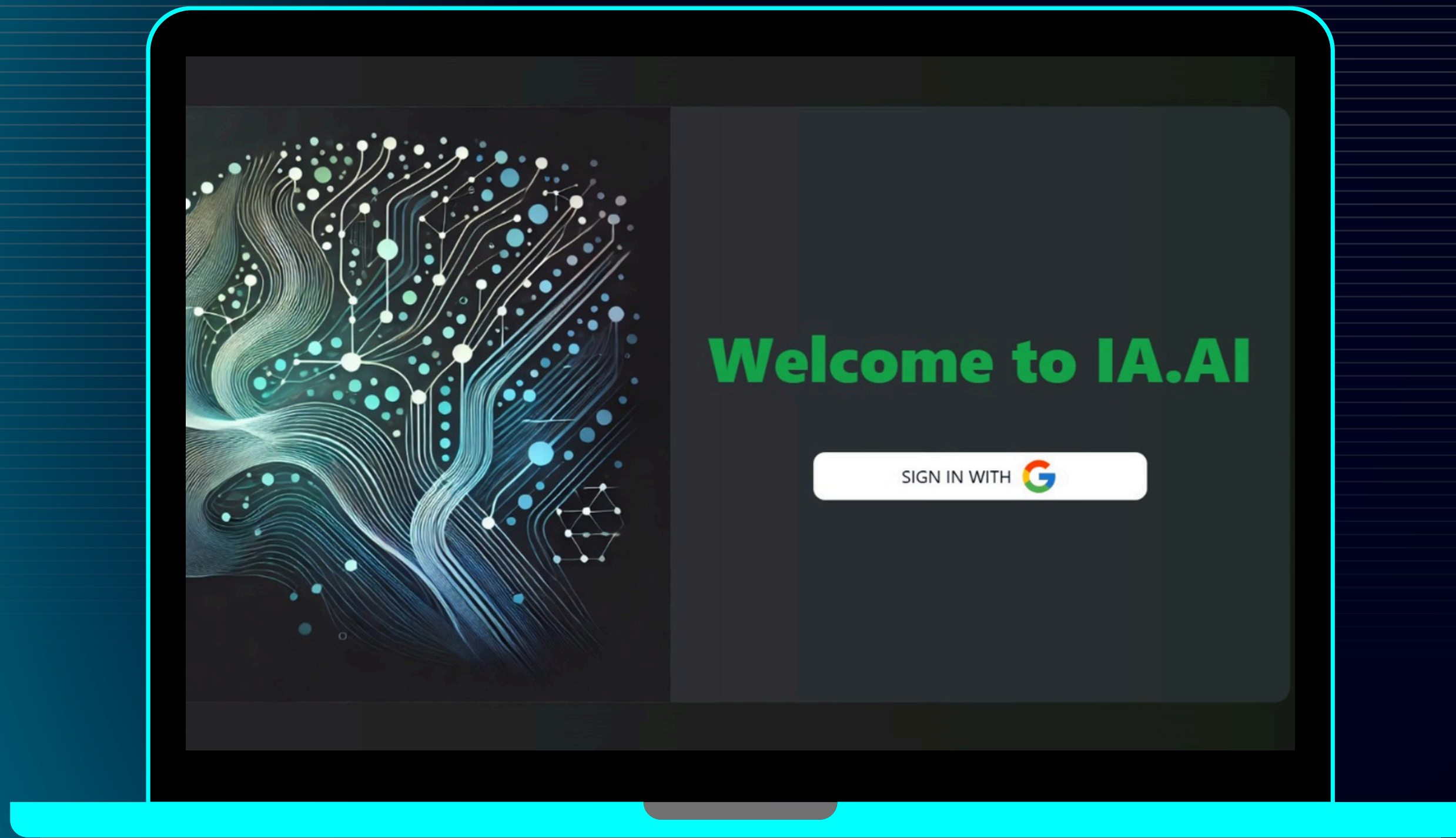
CHROMA VECTOR STORE:

- Stores embeddings for efficient retrieval.

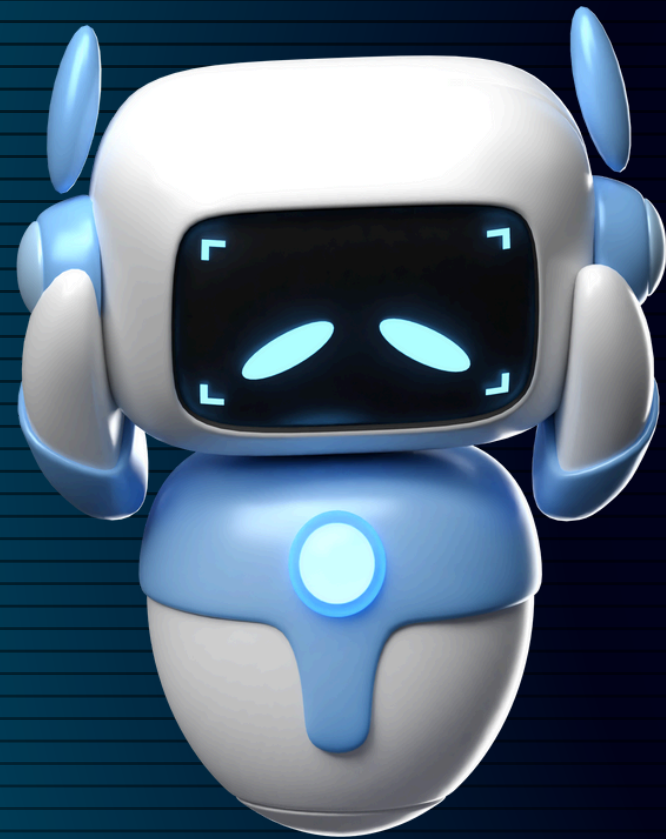
AI SHOULD
COMPLEMENT
TEACHING...



6 DEMO



⑦ CHALLENGES & SOLUTIONS



Challenges:

- a. Real-time voice transcription.
- b. Managing large file uploads.
- c. Optimizing AI response times.

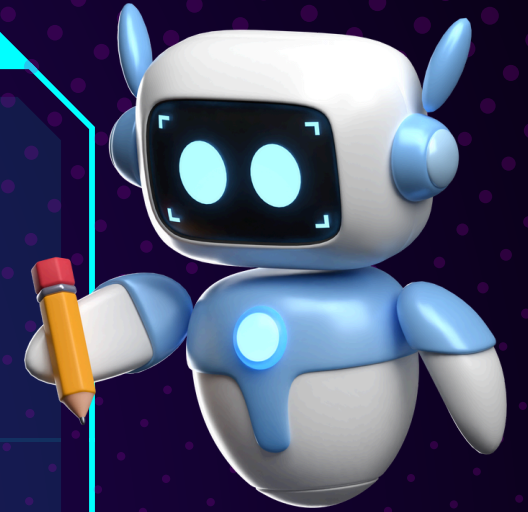
Solutions:

- Used Whisper for efficient transcription.
- Threading for non-blocking processes.
- LangChain for optimized embeddings.

⑧ FUTURE ENHANCEMENTS

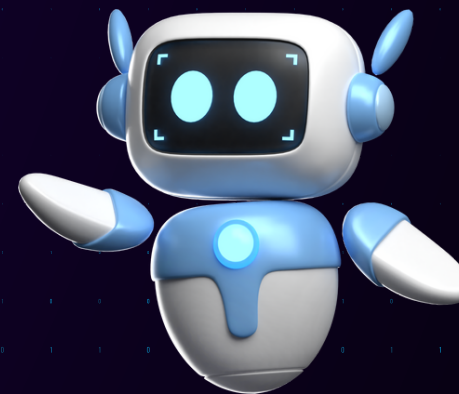
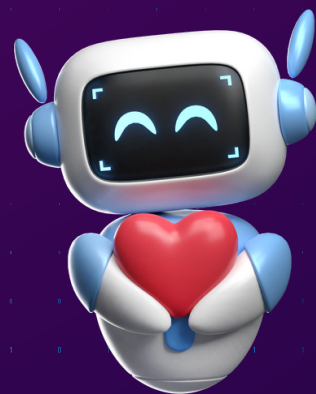
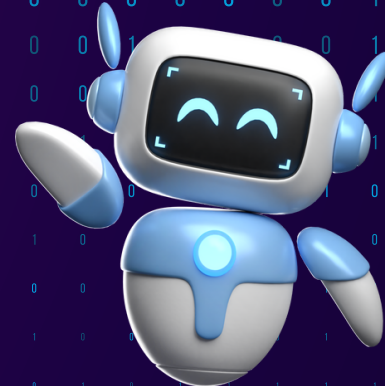
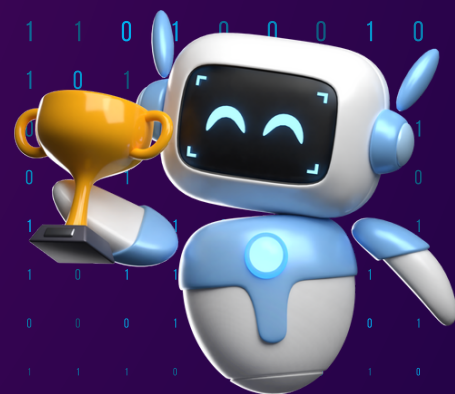
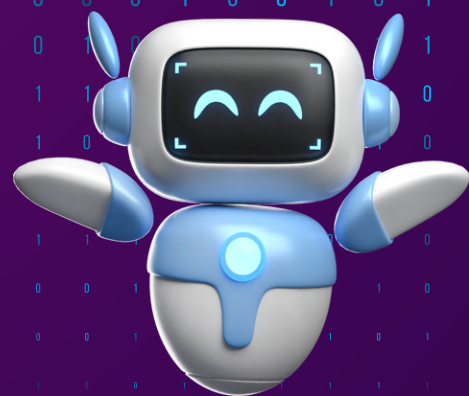


- Enhance AI models for more natural conversations.
- Support additional file formats (e.g., images, videos).
- Improve user authentication for private chats.
- Add Task specific AGENTS .



9 CONCLUSION

- Built a scalable chatbot platform integrating text and voice.
- Seamless interaction between frontend, backend, and AI.
- Demonstrated real-world use cases with advanced AI processing.



THANK YOU ANY
QUESTIONS?

